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1. Bednarek D and Regalla J. **Contagious bovine pleuropneumonia**. *Medycyna Weterynaryjna* 2004; 60: 137-142 (in Polish with an abstract in English).

Address: D. Bednarek, Al Partyzantów 57, 24-100 Pulawy, Poland. E-mail: dbednare@piwet.pulawy.pl

Abstract. The article reviews the etiopathology, epidemiology, clinical manifestations, lesions, diagnosis, and also control and eradication of contagious bovine pleuropneumonia (CBPP), one of the major threats to cattle health and production in Africa and Asia. Considered to have been eradicated several years ago, this disease has now become endemic in the southern countries of Europe. Therefore the epidemic status of contagious bovine pleuropneumonia and its importance for cattle breeding has recently been under consideration, particularly regarding international animal trade and veterinary regulations in the European Union. It is placed in list A of OIE. CBPP is a respiratory disease characterized by croupous pneumonia and serofibrinous pleurisy. The usual clinical form of the disease is acute but chronic forms are also common, especially in endemic regions. Hyperacute forms of CBPP with a high mortality rate can be observed at the beginning of outbreaks in newly infected herds.

2. Bolin SR and Grooms DL. **Origination and consequences of bovine viral diarrhea virus diversity**. *Veterinary Clinics of North America: Food Animal Practice* 2004; 20: 51-68.

Address: D.L. Grooms, Department of Large Animal Clinical Sciences, Michigan State University, A100 VTH, East Lansing, MI 48824, USA. E-mail: groomsd@cvm.msu.edu

Abstract. *Bovine viral diarrhea virus* (BVDV) is characterized by genetic, antigenetic, and pathogenetic diversity. Insight into the diversity of BVDV is important for understanding the wide variety of clinical presentations caused by the virus, for epidemiologic monitoring, and for the optimal design and application of diagnostic tests and vaccines. This reviews focuses on the origin and practical consequences of BVDV diversity.

3. Chase CCL, Elmowalid G and Yousif AAA. **The immune response to bovine viral diarrhea virus: a constantly changing picture**. *Veterinary Clinics of North America: Food Animal Practice* 2004; 20: 95-114.

Address: C.C.L. Chase, Department of Veterinary Science, South Dakota State University, P.O. Box 2175, Brookings, SD 57007, USA. E-mail: christopher chase@sdstate.edu

Abstract. *Bovine viral diarrhea virus* (BVDV) is one of the major immunosuppressive viruses of cattle. The effect on the innate and acquired immune system is unique and results in dramatic immune dysfunction. BVDV infection has also the ability to cause persistent infection (PI) in the developing fetus. This PI syndrome creates a requirement for high levels of BVDV immunity from vaccines to prevent these infections. BVDV vaccines and their future development continue to be an enigma in the control of BVDV.

4. Curtis CF. **Current trends in the treatment of *Sarcoptes*, *Cheyletiella* and *Otodectes* mite infestations in dogs and cats**. *Veterinary Dermatology* 2004; 15: 108-114.

Address: C.F. Curtis, 7 Chadwell, Ware, SG12 9JX, United Kingdom. E-mail: cf.curtis@btopenworld.com

Abstract. For a number of reasons, several of the more 'traditional' ectoparasiticides in the small animal veterinarian's armoury have been withdrawn over the past few years. New, safer products which are long-acting and easier to apply than the conventional dips, rinses and aerosol sprays of the past have replaced them. However, relatively few such novel acaricidal preparations have become commercially available. Consequently, practitioners and researchers frequently experiment with the drugs they have at their disposal to assess their efficacy against a variety of target acarids when used at different dosages and/or via different routes of administration, compared with those recommended by the manufacturer. This paper reviews the anecdotal and peer-reviewed reports describing the use of modern acaricides in dogs and cats that have recently appeared in the veterinary literature. It should be stressed, however, that no medicine should be prescribed for extra-label use without the informed consent of the owner.

5. Duarte ER and Hamdan JS. **Otitis in cattle, an aetiological review.** Journal of Veterinary Medicine, Series B 2004; 51: 1-7.

Address: J.S. Hamdan, Department of Microbiology, Biological Science Institute, Federal University of Minas Gerais, Brazil. E-mail: hamdan@icb.ufmg.br

Abstract. Otitis externa in cattle has a significant impact in tropical and subtropical regions, and the aetiological agents are predominantly rhabditiform nematodes and mites of the genus *Railletia*. Its prevalence is higher in mature and Zebu cattle. In advanced clinical cases there can be irreversible and fatal neural lesions. Ear infection in calves has been associated with concurrent respiratory diseases and mixed infection. The principal reported agents of otitis in calves are bacteria such as *Actinomyces* spp., *Corynebacterium pseudotuberculosis*, *Escherichia coli*, *Haemophilus somnus*, *Pasteurella multocida*, *Mannheimia haemolytica*, *Pseudomonas* spp., *Streptococcus* spp. and *Mycoplasma bovis*. The control and treatment of bovine otitis is not standardized and there is little evidence-based support for the diverse treatments available in the literature.

6. Ewers C, Lübke-Becker A and Wieler LH. ***Mannheimia haemolytica* and the pathogenesis of pneumonic pasteurellosis.** Berliner und Münchener Tierärztliche Wochenschrift 2004; 117: 97-115 (in German with an abstract in English).

Address: C. Ewers, Institut für Mikrobiologie und Tierseuchen, Postfach 040225, 10061 Berlin, Germany

Abstract. *Mannheimia* (M.) *haemolytica* (formerly *Pasteurella* (P) *haemolytica*) is the primary aetiological agent of pneumonic pasteurellosis - one of the most important respiratory diseases in cattle and sheep. While bovine pneumonic pasteurellosis is regarded to be mainly caused by *M. haemolytica* serotype A1, and in Germany during the last years also by serotype A6, sheep can be infected by all serotypes although there is an increased prevalence of serotypes A2 and A5-7. The obligate pathogenicity of *M. haemolytica* is proven by isolation of pure cultures from pneumonic lungs as well as by infection studies. Knowledge about the virulence mechanisms of *M. haemolytica* and their molecular basis are fragmentary, most probably due to the complex gene regulation of virulence associated factors in lung tissues. This review summarizes the current literature covering virulence factors to substantiate a model of pathogenesis. After serotype A1 strains have colonized the bovine upper respiratory tract they replace other serotypes by mechanisms unknown to date. After fulminant proliferation in the upper respiratory tract the microorganisms colonize the lower respiratory tract, finally entering alveolar spaces. An inflammatory cascade is initiated by *M. haemolytica* LPS and Leukotoxin, causing activation of the complement system and release of cytokines. Pathognomonic for bovine pneumonic pasteurellosis is the strong influx of neutrophils accompanied by accumulation of fibrin, finally causing necrosis of alveolar spaces. Depending on lesion size this fibronecrotizing pneumonia can result in death of the animals. In addition, possible protective antigens are discussed. There is still a great effort in the development of efficacious vaccines against pneumonic pasteurellosis in cattle and sheep caused by various *M. haemolytica* serotypes worldwide. The scarce knowledge concerning presence and distribution of virulence associated factors in *M. haemolytica* strains and their role in pathogenesis made it difficult to determine a suitable vaccine candidate in the past. In addition, there is lack of knowledge concerning the variability of virulence factors in individual isolates. Genome sequence analysis of *M. Haemolytica*, enabling proteomics and transcriptomics, hopefully will give new insight into the pathogenesis of pneumonic pasteurellosis.

7. Filippich LJ. **Tumor control in birds.** Seminars in Avian and Exotic Pet Medicine 2004; 13: 25-43.

Address: L.J. Filippich, School of Veterinary Science, University of Queensland, Queensland 4072, Australia

Abstract. Neoplasia is common in pet birds, especially psittacines, and mainly involves the integument and urogenital system. Before treatment options are considered, a definitive diagnosis should be made and the extent of the disease determined. Treatment should initially be directed at tumor eradication and may involve using several modalities together or sequentially. Surgery, radiotherapy, and photodynamic therapy are used against localized tumors, while chemotherapy and biological response modification are

also used against metastatic disease. In combination or adjunct therapy, surgery is used to excise or debulk the tumor, radiotherapy to sterilize local regional disease and chemotherapy and biological therapy to help prevent metastatic disease. The tumor control program should be rationally planned before application, rather than added on when one modality fails, as is commonly practiced. Tumor response to therapy should be regularly assessed both in the short and long term and wherever possible, assessment should be quantitated. Work place health and safety procedures for radiation and cytotoxic drugs should always be practiced.

8. Grooms DL. **Reproductive consequences of infection with bovine viral diarrhea virus.** Veterinary Clinics of North America: Food Animal Practice 2004; 20: 5-19.

Address: D.L. Grooms, Department of Large Animal Clinical Sciences, College of Veterinary Medicine, Michigan State University, A100 VTH, East Lansing, MI 48824, USA. E-mail: groomsd@cvm.msu.edu

Abstract. Bovine viral diarrhea virus (BVDV) has been associated with a wide variety of clinical reproductive problems that are of economic importance including infertility, early embryonic death, abortion, and congenital defects. Fetal infection may also result in the birth of calves that are immunotolerant and persistently infected with BVDV. These calves serve as the major source of virus spread within and between herds. Understanding the reproductive consequences of BVDV is important in properly designing and implementing an effective prevention and control program.

9. Guaguère E, Steffan J and Olivry T. **Cyclosporin A: a new drug in the field of canine dermatology.** Veterinary Dermatology 2004; 15: 61-74.

Address: E. Guaguère, Clinique Vétérinaire Saint Bernard, 598 Avenue de Dunkerque, 59160 Lomme, France. E-mail: eguaguere@nordnet.fr

Abstract. In the last few years, there has been growing interest in the use of cyclosporin to treat canine skin diseases. Cyclosporin exhibits potent immunomodulating properties that reflect its ability to block the transcription of cytokines genes in activated T lymphocytes. Cyclosporin also inhibits a number of immune allergic reactions that occur after activation of mast cells, Langerhans cells, eosinophils and keratinocytes. In randomized controlled trials, cyclosporin has proven to be as effective as glucocorticoids for treatment of canine atopic dermatitis at the inducing dosage of 5 mg kg⁻¹. The drug has also proven beneficial for the treatment of perianal fistulas in dogs. Other potential applications are suggested from small pilot open trials using dogs affected with various immune-mediated dermatological diseases. The pharmacokinetic properties of cyclosporin are very similar in dogs and man, but its safety margin is much wider in dogs. Therefore, routine cyclosporin blood level monitoring does not appear necessary. Although in man renal impairment and hypertension are often seen, even at low doses, these effects are not observed in dogs. Adverse reactions consist mainly of transient emesis and diarrhoea occurring during the first days of treatment. Other adverse reactions, such as gingival hyperplasia, verruciform lesions and hypertrichosis, appear to be dose-dependent, and occur rarely at therapeutic doses. An increased susceptibility to infections has not been reported in dogs receiving this drug.

10. Iff I, Kathmann I, Ledieu D, Allenspach K, Fodor G and Jaggy A. **Seizures in a dog with acute myeloblastic leukaemia: review of literature and case report.** Tierärztliche Praxis, Ausgabe Kleintiere Heimtiere 2004; 32: 15-20 (in German with an abstract in English).

Address: I. Iff, Departement für Klinische Veterinärmedizin der Universität Bern, Postfach, CH-3001 Bern, Switzerland. E-mail: isabelle.iff@kvh.unibe.ch

Abstract. Leukaemia is a neoplastic disease originating from precursor cells of the haematopoietic cells in the bone marrow. According to the FAB (French-American-British-) Classification in humans, leukaemia in animals can be divided into further groups, depending on the results of normal cytology as well as immunohistochemical and cytoenzymatic reactions. Acute leukaemia is a rare disease in dogs. To our knowledge, there have been no descriptions in the veterinary literature on acute leukaemia with concomitant seizures in dogs so far. We therefore reviewed the pertinent literature and describe the case of a four-year-old Dalmatian crossbreed dog referred because of seizures, weakness and incoordination.

According to the results of routine cytology as well as cytoenzymatic and immunohistochemic stainings, the final diagnosis of an acute myeloblastic leukaemia (AML1 or AML5) was confirmed.

11. Kołodziejczyk P and Pejsak Z. **Biological properties of *Mycoplasma hyopneumoniae*, agent of mycoplasmal pneumonia in swine.** Medycyna Weterynaryjna 2004; 60: 341-344 (in Polish with an abstract in English).

Address: Z. Pejsak, Al. Partyzantów 57, 24-100 Pulawy, Poland. E-mail: zpejsak@piwet.pulaway.pl

Abstract. Mycoplasmal pneumonia in swine (MPS) is one of the most common respiratory diseases in this animal. Its characteristic gross lesions in lungs are found in 30%-80% of slaughtered pigs. *Mycoplasma hyopneumoniae* (Mhp), the etiological agent of MPS, is one of the smallest self-replicating bacteria. In contrast to other bacteria from Mycoplasma family it has no cellular wall and this contributes to its considerable polymorphism.

The Mollicutes class contains over 120 Mycoplasma species. Some species are pathogenic for plants, vertebrates and insects. Due to the fact that this bacterium is able to pass through membrane filters of very small diameter in the pores, it has for many years been described as a virus. However, in contrast to viruses, *Mycoplasma* contains both DNA and RNA, is able to grow on a cell-free medium and is sensitive to wide spectrum of antibiotics. Molecular studies have revealed that proteins are located on the surface of cytoplasmatic membrane lipoprotein components like p65, p50 and p44. Through using polyclonal and monoclonal mice, the antibodies of p65 lipoprotein have proven to be the major antigen recognised by the immune system of the macro-organism during infection.

The attachment of Mhp to the host cells is a preliminary and necessary prerequisite for infection. The early stage of infection involves the colonisation of the surface of the trachea, bronchi and bronchioles. Mhp does not enter inside the cells, but locates itself on the top of the cilia. There are structures which play the role of adhesives among the proteins on the Mhp cytoplasmatic membrane and which provide specific binding with the epithelial tissue receptors. The most important adhesive is P97. Infected airways show evidence of ciliary and epithelial cell damage and their antibacterial mucus activity is reduced. The result of this state is catarrhal bronchopneumonia.

Significant immunosuppression is present during Mhp infection. The specific cellular immune response is reduced, contributing to chronic multifactory respiratory tract infections. Thacker and Thacker have proven the significantly negative influence of mixed Mhp and PRRS virus infections in piglets. Bacterial organisms are very fastidious and require specific, enriched media. Cultures grow slowly in primary broth, producing a faint turbidity and an acid colour shift. PCR technique is recommended for the quick detection of Mhp. This method is expensive, so indirect techniques, such as serological examination are the most popular.

12. Kothbauer O. **Anatomical verification of acupuncture in cattle: a review.** Wiener Tierärztliche Monatsschrift 2004; 91: 4-13.

Address: O. Kothbauer, Am Windberg 2, A-4710 Grieskirchen, Austria

Abstract. Researchers of anatomy have searched for principles that would enable recognition of a link between internal organs and the skin. With the benefit of their findings it proved possible to identify important acupuncture points on the skin of cattle in the region of spinal nerve endings, their overlapping zones and their maximum regions. This review presents 12 diagrams based on anatomical topographical research showing acupuncture points and the effects of acupuncture on different organ areas of cattle. The anatomical research has enabled practitioners to recognize sections of acupuncture meridians, assign these according to topographical anatomical criteria and classify acupuncture points in cattle. When sensitive to pressure, acupuncture points (APs) can be used as an additional element in a diagnosis. According to the principles of traditional Chinese veterinary medicine these APs can be used for therapy in their own right, or in addition to other therapeutic methods.

Anatomically defined acupuncture points in the path of acupuncture meridians on the skin of cattle and other animals, as a morphological component with measurable parameters, must be evidenced before they can become an integral part of a holistic system for the maintenance of physiological body

functions. The approach of being prepared through a change of paradigm to work together constructively towards a solution must become reality. Competent acupuncture organizations together with established research teams at the university level are the suitable forum for the research and the practical application of its results.

13. Markine-Goriaynoff N, Minner F, De Fays, Gillet L, Thiry E, Pastoret PP and Vanderplasschen A. **Bovine herpesvirus 4.** *Annales de Médecine Vétérinaire* 2003; 147: 215-247 (in French with an abstract in English).

Address: A. Vanderplasschen, Département des maladies infectieuses et parasitaires, Service d'Immunologie-Vaccinologie, Faculté de Médecine Vétérinaire, Université de Liège, Boulevard de Colonster, 20, bât B43b, 4000 Liège, Belgium. E-mail: a.vdplasschen@ulg.ac.be

Abstract. Bovine herpesvirus 4 (BoHV-4) belongs to the *Herpesviridae* family, *Gammaherpesvirinae* subfamily, *Rhadinovirus* genus like human herpesvirus 8 (HHV-8), the causative agent of Kaposi's sarcoma. BoHV-4 has a worldwide distribution in the cattle population. It has been isolated from cattle showing various clinical signs as well as from healthy cattle. The interest of the scientific community for BoHV-4 is explained by two reasons. Firstly, BoHV-4 represents an homologous virus/host species model to study the biology of gammaherpesviruses. Secondly, the use of BoHV-4 as a recombinant vector for expression both *in vitro* and *in vivo* has been proposed. For these reasons, a considerable amount of data has been collected on this virus. In the present paper, the authors will present a general overview of the literature published on this virus addressing clinical, epidemiological and fundamental aspects of BoHV-4. Finally, in the light of their recent phylogenetic data, the authors will discuss the origin and the host species of BoHV-4 leading to the conclusion that this virus should be considered as a virus of the African buffalo rather than cattle.

14. Messick JB. **Hemotropic mycoplasmas (Hemoplasmas): a review and new insight into pathogenic potential.** *Veterinary Clinical Pathology* 2004; 33: 2-13.

Address: J.B. Messick, University of Illinois at Urbana-Champaign, College of Veterinary Medicine, Urbana, IL 61802, USA. E-mail: jmessick@cvm.uiuc.edu

Abstract. The red cell parasites formerly known as *Haemobartonella* and *Eperythrozoon* spp have been reclassified as hemotropic mycoplasmas (hemoplasmas) based on strong phylogenetic evidence and 16S ribosomal RNA gene sequences. The latter form the basis for polymerase chain reaction assays used to detect infection. *Candidatus* designation was given to incompletely characterized species. Like other mycoplasmas, hemoplasmas are small epicellular parasites that lack a cell wall and are susceptible to tetracyclines; their circular, double-stranded DNA encodes only those gene products essential for life. Diseases caused by infection with hemoplasmas range from overt life-threatening hemolytic anemia to subtle chronic anemia, ill-thrift, and infertility. In addition, the organisms may act as cofactors in the progression of retroviral, neoplastic, and immune-mediated diseases. Intimate contact of hemoplasma organisms with RBCs leads to cell injury through immune-mediated and other mechanisms that have not yet been defined. Despite an intense immune response and even with antibiotic treatment, infected animals probably remain chronic carriers after clinical signs have resolved.

15. Moriello KA. **Treatment of dermatophytosis in dogs and cats: review of published studies.** *Veterinary Dermatology* 2004; 15: 99-107.

Address: K.A. Moriello, Department of Medical Sciences, School of Veterinary Medicine, 2015 Linden Drive West, Madison, WI 53706, USA. E-mail: moriellk@svm.vetmed.wisc.edu

Abstract. The recent literature on the treatment of dermatophytosis in dogs and cats was reviewed. Based upon *in vitro* studies using isolated infected hairs and controlled or field *in vivo* studies, the following topical treatments were consistently found to be antifungal (antidermatophyte): lime sulfur (1:16), 0.2% enilconazole rinses, and a combined 2% miconazole/chlorhexidine shampoo. Animals or hairs were either bathed or rinsed once or twice weekly. Itraconazole, griseofulvin and terbinafine were evaluated in controlled or field studies, most commonly involving cats. Griseofulvin (50 mg kg⁻¹) was

reported to cure infected animals in 41-70 days. Itraconazole (10 mg kg⁻¹ once daily or in a combined daily/pulse therapy 10 mg kg⁻¹ once daily for 28 days and then week on/week off) was reported to cure infected animals in 56-70 days. Low-dose itraconazole (1.5-3.0 mg kg⁻¹) in 15-day cycles required 1-3 cycles (15-45 days). Various doses of terbinafine (5-40 mg kg⁻¹) were reportedly used to treat dogs or cats. The higher doses of terbinafine (>20 mg kg⁻¹) were required to achieve a mycological cure; the number of treatment days to cure varied from 21 to >126 days. Lufenuron was reported anecdotally to be an effective cure, however, this was not substantiated in controlled studies. Finally, fungal vaccines were not found to be effective against challenge exposure, however, there is evidence that they may be useful in treatment protocols.

16. Mueller RS. **Treatment protocols for demodicosis: an evidence-based review.** *Veterinary Dermatology* 2004; 15: 75-89.

Address: R.S. Mueller, Department of Clinical Sciences, College of Veterinary Medicine and Biomedical Sciences, Colorado State University, Fort Collins, CO 80523, USA. E-mail: rmuller@colostate.edu

Abstract. Publications discussing the treatment of demodicosis in the dog and cat are reviewed. Based on the evidence in the literature, amitraz rinses at 0.025-0.06% every 7-14 days, and oral daily ivermectin at 300 µg kg⁻¹, milbemycin at 2 mg kg⁻¹ and moxidectin at 400 µg kg⁻¹, respectively, can all be recommended for the treatment of generalized canine demodicosis. Ivermectin and moxidectin should be initiated at lower doses and patients monitored for possible adverse effects during therapy. In cats, 2% lime sulfur dips and amitraz rinses at 0.0125-0.025% have been used successfully.

17. Niedbalski W and Kęsy A. **Foot-and-mouth disease - the carrier state in ruminants.** *Medycyna Weterynaryjna* 2004; 60: 115-118 (in Polish with an abstract in English).

Address: W. Niedbalski, ul Zielona 48/4, 98-220 Zduńska Wola, Poland. Email: wieslaw@piwzp.invar.net.pl

Abstract. After contact with the foot-and-mouth disease virus (FMDV) ruminants may become chronically infected (i.e. carrier animals). These animals are a potential source of infection and can play a role in the epidemiology of foot-and-mouth disease (FMD). Carrier animals are defined as animals from which FMDV can be isolated from oropharyngeal fluid (OPF) more than 28 days post infection. Chronic infection can occur either after clinical or subclinical FMD infection, and in vaccinated and non-vaccinated animals. The detection of carrier animals has resulted in international trade restrictions. The carrier state is also an important barrier in using prophylactic vaccines against foot-and-mouth disease. This review describes the occurrence of carrier animals and available methods of detection. It also presents the location and properties of the chronic virus as well as the mechanisms of chronic infection and the possible role of carriers in transmitting FMDV.

18. Otrócka-Domagala I, Słodki S and Rotkiewicz T. **Immunophenotypic variability of leukocytes in cattle infected with BLV.** *Medycyna Weterynaryjna* 2004; 60: 249-253 (in Polish with an abstract in English).

Address: I. Otrócka-Domagala, ul Jarocka 72/12, 10-699 Olsztyn, Poland. E-mail: tadrot@moskit.uwm.edu.pl

Abstract. The article presents the results of research done to date on the variability of expression of superficial markers on leukocytes in cattle infected with *bovine leukemia virus* (BLV). Studies carried out so far show that these changes mainly concern B lymphocytes, and, to a lesser degree monocytes, macrophages and T lymphocytes. The article also describes the role of flow cytometry as a quick and precise method for assessing immunophenotype cells in the process of leukemia in cattle.

19. Prélard P. **Clinical diagnosis of allergic skin diseases in the dog.** *Revue de Médecine Vétérinaire* 2004; 155: 12-19.

Address: P. Prélard, Cabinet de dermatologie vétérinaire, 62 Avenue Maurice Clétras, 44300 Nantes, France

Abstract. It is possible to make a clinical diagnosis of urticaria, canine atopic dermatitis (CAD), of flea allergy dermatitis (FAD) without specific diagnostic tests. Such a diagnosis is based on history (age of

onset for or steroid sensitive pruritus for CAD) and clinical signs (type of lesions for urticaria, localization of lesions for CAD, FAD and contact dermatitis) and eviction of ectoparasites other than fleas as a cause of pruritus. Allergy testing - skin tests, eviction diet, IgE dosages - are only used for etiological diagnosis.

20. Rosenkrantz WS. **Pemphigus: current therapy.** *Veterinary Dermatology* 2004; 15: 90-98.

Address: W.S. Rosenkrantz, 2965 Edinger Ave, Tustin, CA 92780, USA. E-mail: AnDerm1@aol.com

Abstract. Pemphigus is an autoimmune skin disease that can present in a variety of forms and can be a challenging disease to manage and treat. An overview of the different forms of pemphigus and diagnostics are discussed including pemphigus foliaceus (PF), pemphigus erythematosus (PE), panepidermal pustular pemphigus (PPP), pemphigus vulgaris (PV) and paraneoplastic pemphigus (PNP). Emphasis on therapy is presented. Included are the most current commonly used therapeutics (glucocorticoids, azathioprine, chlorambucil and tetracycline and niacinamide); current alternative therapeutics (cyclosporin and tacrolimus and mycophenolate mofetil) and additional alternative therapeutics (cyclophosphamide, chrysotherapy, dapsone, sulfasalazine and intravenous immunoglobulin (IVIG) therapy).

21. Sobczyńska-Rak A and Różańska D. ***Helicobacter* in animals and people.** *Medycyna Weterynaryjna* 2004; 60: 132-136 (in Polish with an abstract in English).

Address: A. Sobczyńska-Rak, ul Gleboka 30, 20-612 Lublin, Poland. E-mail: olsob@poczta.onet.pl

Abstract. The presence of *Helicobacter* sp. in animal stomachs has been described since 1881. In people infection caused by *Helicobacter pylori* always evokes the inflammation of stomach mucous and it appears in more than 95% of patients affected by duodenum ulcers and in 75% of patients with gastric ulcers.

The presence of *H. helis*, *H. heilmannii*, *H. bizzozeronii*, *H. billis* was confirmed in dog stomachs and in cats, *H. felis*, *H. heilmannii*, *H. pylori*. Gastritis is a concomitant disease in only part of the dog and cat populations. In most of the animals the infection does not evoke any clinical symptoms. Occasionally medium to severe gastritis occurs, characterised by paroxysmal vomiting or diarrhoea. The diagnostic methods can be divided into the invasive and non-invasive methods. The effective elimination of the *Helicobacter* infection is possible by combined therapy, using gastric inhibition secretion drugs and at least two antibiotics.

22. Turek MM. **Cutaneous paraneoplastic syndromes in dogs and cats: a review of the literature.** *Veterinary Dermatology* 2003; 14: 279-296.

Address: M.M. Turek, School of Veterinary Medicine, Department of Surgical Sciences, University of Wisconsin-Madison, 2015 Linden Drive Madison, WI 53706, USA. E-mail: turekm@svm.vetmed.wisc.edu

Abstract. Cutaneous paraneoplastic syndromes are a group of noncancerous dermatoses associated with internal malignancy. Their recognition can facilitate detection and timely treatment of underlying cancer. More than 30 such disorders have been identified in the human scientific literature, whereas only a few are described in veterinary medicine. This may reflect a lower incidence in animals than in people or may be the result of failure to recognize an association between certain skin lesions and neoplasia. Establishing a relationship between a cutaneous disorder and neoplasia can be difficult unless the skin lesions are rare and almost always associated with a particular tumour type, as is the case for most recognized veterinary paraneoplastic dermatoses. Among these are feline paraneoplastic alopecia, feline thymoma-associated exfoliative dermatitis, nodular dermatofibrosis, feminization syndrome associated with testicular tumours, superficial necrolytic dermatitis and paraneoplastic pemphigus. The aetiology of most cutaneous paraneoplastic syndromes has remained elusive in both people and animals.

23. Wilkins PA and Seahorn T. **Acute respiratory distress syndrome.** *Veterinary Clinics of North America: Equine Practice* 2004; 20: 253-273.

Address: P.A. Wilkins, Department of Clinical Studies, University of Pennsylvania School of Veterinary Medicine, New Bolton Center, 382 West Street Road, Kennett Square, PA 19348, USA. E-mail: pwilkins@vet.upenn.edu

Abstract. Acute respiratory distress syndrome (ARDS) and the less severe acute lung injury (ALI) have been the subject of much clinical and basic research efforts since they were first described in humans in the mid-1960s. The veterinary patient is also affected by ARDS and ALI, although investigations of ARDS and ALI in veterinary medicine are not common. This article reviews some of what is known regarding ARDS and ALI in human and veterinary medicine to both acquaint the unfamiliar reader with the syndromes and update veterinary clinicians seeing these cases in practice. Much work remains to be done regarding understanding the pathophysiology of these syndromes and developing effective therapies.

24. Wlaschitz S. **Grass sickness: a literature review.** *Wiener Tierärztliche Monatsschrift* 2004; 91: 14-22 (in German with an abstract in English).

Address: S. Wlaschitz, Veterinärplatz 1, A-1210 Wien, Austria. E-mail: sonja.wlaschitz@vu-wien.ac.at

Abstract. This paper offers a comprehensive literature review in regard of equine grass sickness (GS). The following topics will be discussed: epidemiological risk factors like age, access to pasture, stress or endoparasites; possible etiologies like a toxicoinfection with *Clostridium botulinum* type C neurotoxin as well as toxins from *Fusarium* species serving as a trigger; clinical signs of the acute, subacute and chronic form; light and electron microscopical neuronal lesions; diagnostic possibilities as contrast radiography of the oesophagus, phenylephrine eyedrop test or ileal biopsy and therapeutic options in the treatment of chronic GS with the aid of nutritional management and prokinetic drugs.

25. Wyse CA, Preston T, Yam PS, Sutton DGM, Christley RM, Hotchkiss JW, Mills CA, Glidle A, Cumming DRS, Cooper JM and Love S. **Current and future uses of breath analysis as a diagnostic tool.** *Veterinary Record* 2004; 154: 353-360.

Address: C.A. Wyse, Institute of Comparative Medicine, University of Glasgow Veterinary School, Bearsden, Glasgow G61 1QH, Scotland

Abstract. The analysis of exhaled breath is a potentially useful method for application in veterinary diagnostics. Breath samples can be easily collected from animals by means of a face mask on chamber with minimal disturbance to the animal. After the administration of a ^{13}C -labelled compound the recovery of ^{13}C in breath can be used to investigate gastrointestinal and digestive functions. Exhaled hydrogen can be used to assess oro-caecal transit time and malabsorption, and exhaled nitric oxide, carbon monoxide and pentane can be used to assess oxidative exhaled breath condensate) can be used to assess airway inflammation. This review summarises the current status of breath analysis in veterinary medicine, and analyses its potential for assessing animal health and disease.